MTH1002-B

UNIVERSITY OF EXETER

FACULTY OF ENVIRONMENT, SCIENCE AND ECONOMY

MATHEMATICS AND STATISTICS

May 2024

Mathematical Methods

Module Leader: Dr Layal Hakim

Duration: 2 HOURS

Answer Section A (50%) and any TWO of the three questions in Section B (25% for each).

Marks shown in questions are merely a guideline. Candidates are permitted to use approved portable electronic calculators in this examination.

This is a **CLOSED NOTE** examination.

SECTION A

1. (a) Let

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}.$$

Determine whether the inverse of A exists, and if it does, find the inverse using Gaussian elimination. (8)

(b) Find and classify the critical point of the function

$$f(x, y) = x^3 y + \frac{1}{2} y^2 - 3x.$$

(8)

(c) Find the eigenvalues and eigenvectors of the matrix

$$A = \begin{pmatrix} 5 & 3 \\ 7 & 1 \end{pmatrix}.$$

(8)

(d) Express the definite integral

$$\int_0^2 (4x + 7)\, dx$$

as a Riemann sum and evaluate it. Verify your answer by calculating the integral using the rules of integration. (6)

(e) Evaluate the following integrals.

(i)

$$\int \frac{\sin(x)}{\cos^3(x)}\, dx$$

(ii)

$$\int x^2 \ln(x)\, dx$$

(10)

(f) Solve the ordinary differential equation

$$\frac{dy}{dx} = \frac{y^2 - 1}{x}$$

with the initial condition $y(1) = 2$. (10)

[**50**]

SECTION B

2. (a) Find the Taylor series expansion of the function

$$f(x, y) = x\cos(xy)$$

about $(x, y) = (\pi, 0)$ up to and including the quadratic term. (10)

(b) Let $u = f(x, y), x = r\cos(\theta), y = r\sin(\theta)$.

(i) Show that

$$u_{rr} = \cos^2(\theta) f_{xx} + 2\cos(\theta)\sin(\theta) f_{xy} + \sin^2(\theta) f_{yy}$$

and

$$u_{\theta\theta} + r u_r = r^2 \left(\sin^2(\theta) f_{xx} - 2\cos(\theta)\sin(\theta) f_{xy} + \cos^2(\theta) f_{yy}\right).$$

(ii) Hence, show that

$$f_{xx} + f_{yy} = u_{rr} + \frac{1}{r^2} u_{\theta\theta} + \frac{1}{r} u_r.$$

(15)

[**25**]

3. (a) Let $f(x, y) = x - y$, and define D as the region above the x-axis bounded by $y^2 = 3x$ and $y^2 = 4 - x$.

(i) Sketch the region D.

(ii) Evaluate

$$I = \int_D f(x, y)\, dxdy.$$

(13)

(b) Find the point on the surface $z = xy - 1$ that is nearest the origin.

(12)

[**25**]

4. (a) For functions $f(x, y, z)$ and $g(x, y, z)$, prove

$$\nabla(fg) = f\nabla(g) + g\nabla(f).$$

(4)

(b) Consider a system of the linear equations

$$\begin{pmatrix} 1 & 1 & 1 \\ a & 0 & 2 \\ 3 & a & 5 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} a \\ 3 \\ 7 \end{pmatrix}.$$

Determine the values of a such that the system has (i) no solution, (ii) a unique solution and (iii) an infinite number of solutions. (12)

(c) (i) For $x = r\cos(\theta)$ and $y = r\sin(\theta)$, calculate

$$\left| \det \begin{pmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{pmatrix} \right|.$$

(ii) Evaluate the integral

$$I = \iint_D y \, dxdy$$

where D is the region outside the circle $r = 1$ and inside the cardioid $r = 1 + \cos(\theta)$. (9)

[**25**]